

Software Development Fundamentals Lab – I [24B15CS111]

Evaluation-1

Monday 14/09/2026 10.00:12.00 pm

ODD MACHINE

Time: 50 Minutes

Max Marks: 15

Q1- Write a C program by using switch case statement:

(8 marks)

Write a C program to simulate an ATM Transaction System. Assume the initial balance is ₹25,000. Display:

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Mini Statement
5. Exit

Deposit amount must be positive. Withdrawal is allowed only if sufficient balance exists. A single withdrawal above ₹10,000 must be rejected. If balance falls below ₹2,000, display a warning. Maintain the number of successful deposits and withdrawals. Use switch-case for menu selection. Use do-while to repeatedly display the menu.

Sample Input	Sample Output
==== ATM MENU ===== 1. Check Balance 2. Deposit Money 3. Withdraw Money 4. Mini Statement 5. Exit Enter your choice: 2 Enter deposit amount: 5000 Enter your choice: 3 Enter withdrawal amount: 7000 Enter your choice: 1	Amount Deposited Successfully. Current Balance: Rs. 30000 Withdrawal Successful. Current Balance: Rs. 23000 Current Balance: Rs. 23000

Q2- Write a C program by using loops- if- else, for, while and do while.

(7 marks)

An e-commerce company wants to analyze customer orders. Take Number of customers N, Customer ID, Number of products purchased, Price of each product. Consider:

Order Value	Discount
₹10,000 or more	20%
₹5,000–₹9,999	10%
Below ₹5,000	No discount

If Final amount > ₹8,000 → Free Delivery Otherwise → ₹100 delivery charge

Also display- Final payable amount for every customer, Customer with the highest purchase amount and Total revenue.

Sample Input	Sample Output
Enter number of customers: 3	===== ORDER SUMMARY =====
Enter Customer ID: 101	Customer ID: 101
Enter number of products: 3	Order Value : Rs. 5500
Enter price of product 1: 2500	Discount : Rs. 550
Enter price of product 2: 1800	Delivery Charge : Rs. 100
Enter price of product 3: 1200	Final Amount : Rs. 5050
Enter Customer ID: 102	Customer ID: 102
Enter number of products: 2	Order Value : Rs. 11000
Enter price of product 1: 6000	Discount : Rs. 2200
Enter price of product 2: 5000	Delivery Charge : Free
Enter Customer ID: 103	Final Amount : Rs. 8800
Enter number of products: 2	Customer ID: 103
Enter price of product 1: 1500	Order Value : Rs. 3500
Enter price of product 2: 2000	Discount : Rs. 0
	Delivery Charge : Rs. 100
	Final Amount : Rs. 3600
	=====
	Highest Purchase Customer : 102
	Highest Order Value : Rs. 11000
	Total Revenue : Rs. 17450

EVEN MACHINE

Q1- Write a C program by using switch case statement:

(8 marks)

A shopping mall wants to develop a Smart Parking Management System. The parking charges depend on the type of vehicle and the number of hours parked.

Display the following menu:

1. Car
2. Bike
3. SUV
4. Exit

The parking charges are:

Vehicle	First 2 Hours	Every Additional Hour
Bike	₹20	₹10
Car	₹50	₹20
SUV	₹80	₹30

If parking exceeds 8 hours, an additional ₹100 penalty is charged. If hours are less than or equal to 0, display Invalid Duration. After generating the bill, display the menu again. The program terminates when the user selects Exit.

Sample Input ===== SMART PARKING SYSTEM ===== 1. Car 2. Bike 3. SUV 4. Exit Enter your choice: 1 Enter parking hours: 5	Sample Output Vehicle Type : Car Parking Duration : 5 hours Base Charge : Rs. 110 Penalty : Rs. 0 Total Parking Fee : Rs. 110 Thank you for using Smart Parking System.
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Q2- Write a C program by using loops- if- else, for, while and do while.
(7 marks)

A digital wallet company wants to identify potentially risky transactions. Accept transaction amounts repeatedly. Classification of transaction-

Transaction Amount	Category
$\leq ₹0$	Invalid
₹1–₹2,000	Normal
₹2,001–₹10,000	Medium Risk
₹10,001–₹50,000	High Risk
$> ₹50,000$	Suspicious

If three consecutive transactions are above ₹10,000, display Account Temporarily Blocked. If total transaction amount exceeds ₹1,00,000, stop accepting transactions. Count each category. Display total transaction amount and category-wise counts.

Sample Input: Enter transaction amount: 1500 Enter transaction amount: 7500 Enter transaction amount: 15000 Enter transaction amount: 22000 Enter transaction amount: 12000	Sample Output: Transaction: Rs. 1500 Category: Normal Transaction Transaction: Rs. 7500 Category: Medium Risk Transaction: Rs. 15000 Category: High Risk Transaction: Rs. 22000 Category: High Risk Transaction: Rs. 12000 Category: High Risk WARNING: Three consecutive high-risk transactions detected! ACCOUNT TEMPORARILY BLOCKED. ===== TRANSACTION SUMMARY ===== Total Transaction Amount : Rs. 58000 Normal Transactions : 1 Medium Risk Transactions : 1 High Risk Transactions : 3 Suspicious Transactions : 0 Transaction processing terminated.
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